

Are autoimmune diseases predictable?

Autoimmune diseases are complex diseases resulting of the interaction between both genetics and environmental factors over time. Different phases in the development of autoimmune diseases are characterized by the detection of serum autoantibodies several months or years before the onset of clinical manifestations and subsequent diagnosis. In addition to serum antibodies, genetic susceptibility factors may predict the future development of the disease. Currently, prediction in type 1 diabetes is the most accurate, with the analysis of genetic susceptibility factors in first-degree relatives of patients and several autoantibody tests. In the future, multiple antibodies test, in combination with the analysis of genetics, epigenetics and immunological anomalies in fine models may allow the precise prediction in autoimmune diseases. Prevention measures might thus be introduced as an attempt to avoid or delay the disease. Copyright © 2011 Elsevier B.V. All rights reserved.